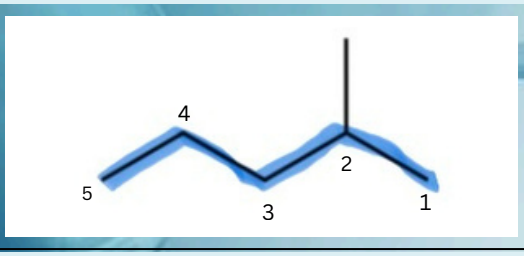
CARD A01 · STRUCTURE ➡ NAME

FOUNDATION ⚫ MOLECULAR FORMULA: C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 5 C pentane. Several possible 5-C paths exist, all carrying a methyl group.

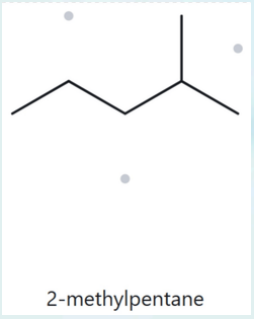
Numbering: left right gives 2 and right left gives 4. First point of difference: 4 > 2, so number from left to right.

Substituents: methyl on C2 → 2-methyl.

3. My Name (Written First)

2-methylpentane

4. The Alkane Check



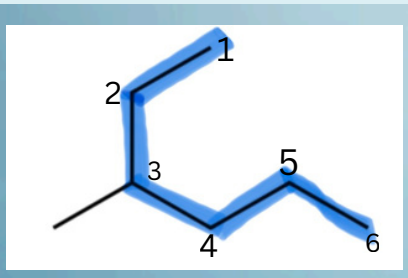
✔Match

✔longest chain confirmed ✔lowest locants ✔alphabetical order ✔punctuation ✔formula counted

CARD A02 · STRUCTURE ➡ NAME

FOUNDATION ⚫ MOLECULAR FORMULA: C_7H_{16}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 6 Chexane. More than one possible 6-C paths exist, but all carry a methyl group.

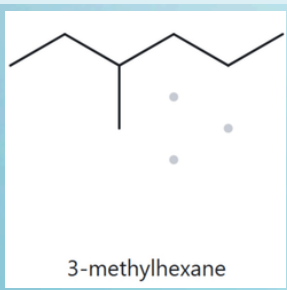
Numbering: top bottom right gives 3 and bottom left → top gives 4. First point of difference: 4 > 3, so number from top to bottom right.

Substituents: methyl on C3 3-methyl.

3. My Name (Written First)

4-methylhexane

4. The Alkane Check



✖Mismatch

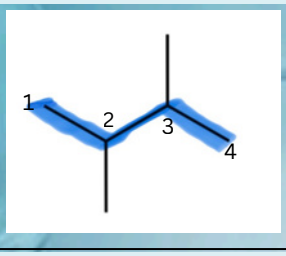
IUPAC rules require starting from the end that yields the lower locant (3 < 4), making 3-methylhexane correct.

✔longest chain confirmed ✔lowest locants ✔alphabetical order ✔punctuation ✔formula counted

CARD A03 · STRUCTURE ➡️ NAME

FOUNDATION ⚫ MOLECULAR FORMULA: C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 4 C → butane. More than one possible 4-C paths exist, but both carry a methyl group so they lead to the same name.

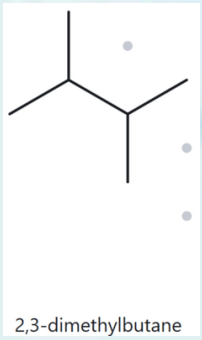
Numbering: left → right gives 2,3 and right → left also gives 2,3. Either way, both are numbered the same.

Substituents: methyl on C2 and C3 2,3-dimethyl.

3. My Name (Written First)

2, 3-dimethylbutane

4. The Alkane Check



2,3-dimethylbutane

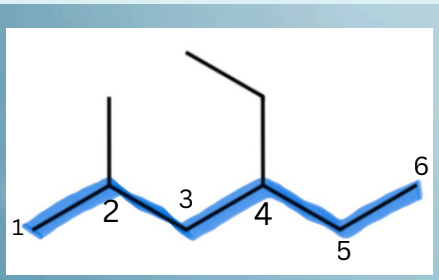
✔️Match

✔️longest chain confirmed ✔️lowest locants ✔️alphabetical order ✔️punctuation ✔️formula counted

CARD A04 · STRUCTURE ➡️ NAME

DEVELOPING ⚫ MOLECULAR FORMULA: C_9H_{20}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 6 Chexane. Several possible 6-C paths exist, but all carry the same methyl and ethyl group so they lead to the same name.

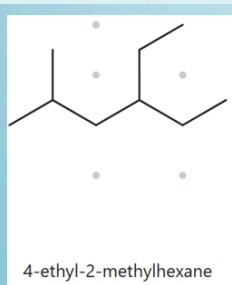
Numbering: left→ right gives 2,4 and right left gives 3,5. We choose 2,4 because it is the lower set at the first point of difference. Therefore, number from left to right.

Substituents: methyl on C2 and ethyl on C4→ 4-ethyl-2-methyl.

3. My Name (Written First)

2-methyl-4-ethylhexane

4. The Alkane Check



4-ethyl-2-methylhexane

✖️Mismatch

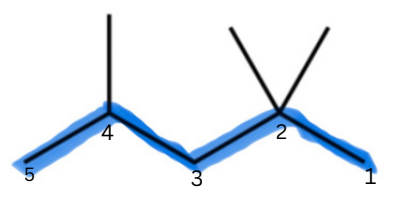
IUPAC rules require citing "ethyl" before "methyl" regardless of their position numbers, making 4-ethyl-2-methylhexane correct.

✔️longest chain confirmed ✔️lowest locants ✔️alphabetical order ✔️punctuation ✔️formula counted

CARD A05 · STRUCTURE ➡️ NAME

DEVELOPING ⚫ MOLECULAR FORMULA: C_8H_{18}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 5 C pentane. Several possible 5-C paths exist, but all carry 3 methyl groups.

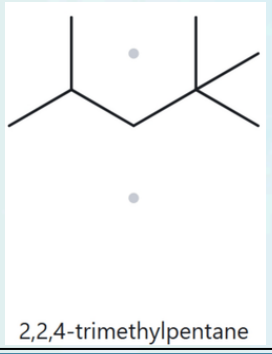
Numbering: left right gives 2,4,4 and right → left gives 2,2,4. 4 > 2; 2,2,4 is lower at the first point of difference. Therefore, number from right to left.

Substituents: 2 methyl on C2 and one on C4 → 2,2,4-trimethyl

3. My Name (Written First)

2,2,4-trimethylpentane

4. The Alkane Check



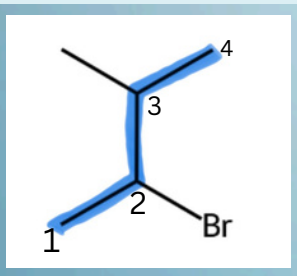
✔️match

✔️longest chain confirmed ✔️lowest locants ✔️alphabetical order ✔️punctuation ✔️formula counted

CARD A06 · STRUCTURE ➡️ NAME

DEVELOPING ⚫ MOLECULAR FORMULA: $C_5H_{11}Br$

1. Numbered Parent



2. Naming Trail

Parent: longest chain 4 C butane. More than one 4-C paths exist, but both carry a methyl and bromo group.

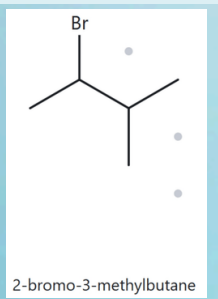
Numbering: bottom top gives 2,3 and top bottom also gives 2,3. Since both give the same set of numbers (2, 3), the tie is broken alphabetically: bromo comes before methyl, so bromo gets the lower number. Therefore, number from lower left-end to top.

Substituents: methyl on C3 and a bromo substituent on C2 → 2-bromo-3-methyl

3. My Name (Written First)

2-bromo-3-methylbutane

4. The Alkane Check



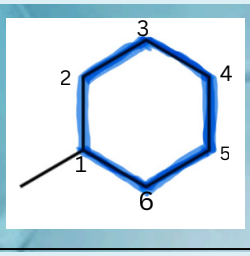
✔️match

✔️longest chain confirmed ✔️lowest locants ✔️alphabetical order ✔️punctuation ✔️formula counted

CARD A07 · STRUCTURE ➡ NAME

DEVELOPING ⚫ MOLECULAR FORMULA: C_7H_{14}

1. Numbered Parent



2. Naming Trail

Parent: longest chain ring with 6 C cyclohexane. It carries a methyl group.

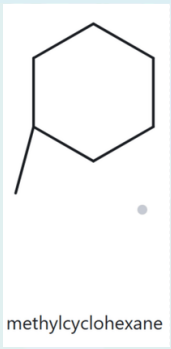
Numbering: There is no need to assign a number because there's only one substituent on the ring. That carbon is automatically C-1.

Substituents: methyl on C1 (but there's only one substituent on the ring; therefore, "1-" before methyl is unnecessary.) → methyl

3. My Name (Written First)

methylcyclohexane

4. The Alkane Check



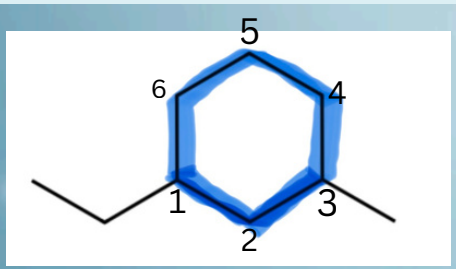
✔match

✔longest chain confirmed ✔lowest locants ✔alphabetical order ✔punctuation ✔formula counted

CARD A08 · STRUCTURE ➡ NAME

CHALLENGE ⚫ MOLECULAR FORMULA: C_9H_{18}

1. Numbered Parent



2. Naming Trail

Parent: longest chain ring with 6 C cyclohexane. It carries a methyl and an ethyl group.

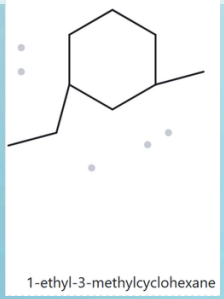
Numbering: Both directions give 1,3, but the substituent that comes first alphabetically (ethyl) gets C-1. Therefore, number from the ethyl group and counterclockwise.

Substituents: ethyl on C1 and methyl on C3 1-ethyl-3-methyl

3. My Name (Written First)

1-ethyl-3-methylcyclohexane

4. The Alkane Check



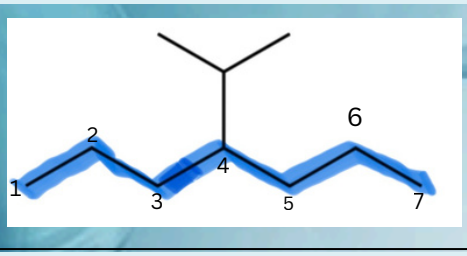
✔match

✔longest chain confirmed ✔lowest locants ✔alphabetical order ✔punctuation ✔formula counted

CARD A09 · STRUCTURE ➡ NAME

CHALLENGE ⚫ MOLECULAR FORMULA: $C_{10}H_{22}$

1. Numbered Parent



2. Naming Trail

Parent: longest chain 6 Chexane. It carries a methyl and propyl group.

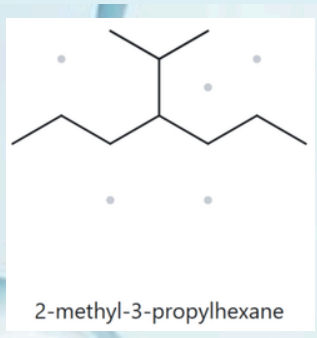
Numbering: top lower-end right gives 2,3 and top → lower-end left also gives 2,3. Therefore, we can number from either side and it will still be the same.

Substituents: methyl on C2 and propyl on C3 2-methyl-3-propyl

3. My Name (Written First)

3-propylheptane

4. The Alkane Check



✖Mismatch

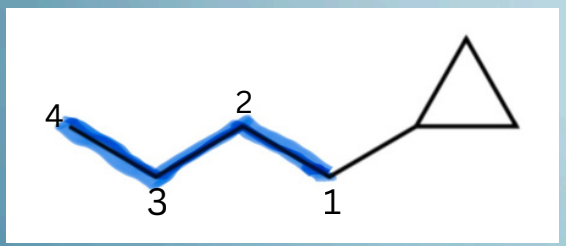
3-propylheptane stops tracing at a branch point instead of following the full continuous 9-carbon path. IUPAC rules require identifying the longest continuous chain (\bm{\text{nonane}}), making 4-methylnonane correct.

✔longest chain confirmed ✔lowest locants ✔alphabetical order ✔punctuation ✔formula counted

CARD A10 · STRUCTURE ➡ NAME

CHALLENGE ⚫ MOLECULAR FORMULA: C_8H_{16}

1. Numbered Parent



2. Naming Trail

Parent: 3-carbon ring cyclopropane. It carries a 4-carbon chain substituent. The longest-chain rule does not override the ring-parent. Therefore, the ring is the parent.

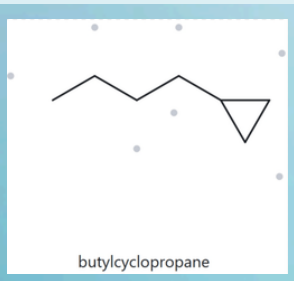
Numbering: The carbon of the cyclopropane ring where the butyl chain attaches is considered C1. However, because there's only one substituent on the ring, there is no need to include the locant in the name.

Substituents: butyl (A locant is unnecessary because there is only one substituent on the cyclopropane ring.) → butyl

3. My Name (Written First)

1-cyclopropylpentane

4. The Alkane Check



✖Mismatch

1-cyclopropylpentane selects the 5-carbon open chain as the parent instead of the ring. Per the 2013 IUPAC rules used in this activity, any ring present is designated as the parent, making pentylcyclopropane correct.

✔longest chain confirmed ✔lowest locants ✔alphabetical order ✔punctuation ✔formula counted

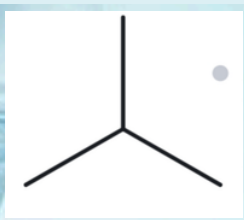
CARD B01 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 2 – methylpropane

1. Name Decoded

Parent: Propane → a 3-carbon chain.
Substituents: Methyl on C2.
Expected formula: 3 + 1 = 4 carbons; open chain → C_4H_{10}

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_4H_{10}.
Open-chain formula: C_nH_{2n+2} = C_4H_{10}. ✓

4. Back-Check

Returned by The Alkane: 2-methylpropane.

✓ The returned name matches the given name.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

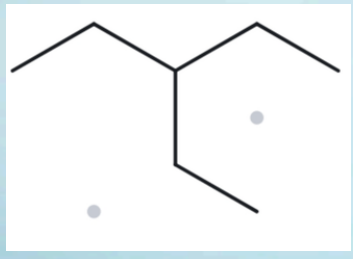
CARD B02 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 3 – ethylpentane

1. Name Decoded

Parent: Pentane → a 5-carbon chain.
Substituents: Ethyl on C3.
Expected formula: 5 + 2 = 7 carbons; open chain → C_7H_{16}

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_7H_{16}.
Open-chain formula: C_nH_{2n+2} = C_7H_{16}. ✓

4. Back-Check

Returned by The Alkane: 3-ethylpentane.

✓ The returned name matches the given name.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

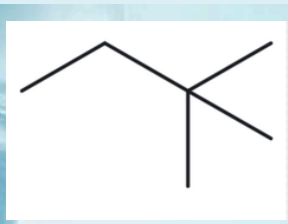
CARD B03 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 2, 2 – dimethylbutane

1. Name Decoded

Parent: Butane → a 4-carbon chain.
Substituents: Two methyl groups on C2.
Expected formula: 4 + 2 = 6 carbons; open chain → C_6H_{14}

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_6H_{14}.
Open-chain formula: C_nH_{2n+2} = C_6H_{14}. ✓

4. Back-Check

Returned by The Alkane: 2,2-dimethylbutane.

✓ The returned name matches the given name.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

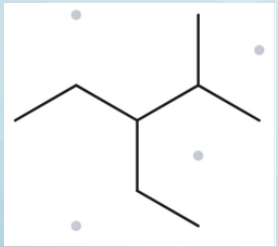
CARD B04 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 3 – ethyl-2-methylpentane

1. Name Decoded

Parent: Pentane → a 5-carbon chain.
Substituents: Ethyl on C3; methyl on C2.
Expected formula: 5 + 2 + 1 = 8 carbons; open chain → C_8H_{18}

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_8H_{18}.
Open-chain formula: C_nH_{2n+2} = C_8H_{18}. ✓

4. Back-Check

Returned by The Alkane: 3-ethyl-2-methylpentane.

✓ The returned name matches the given name.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

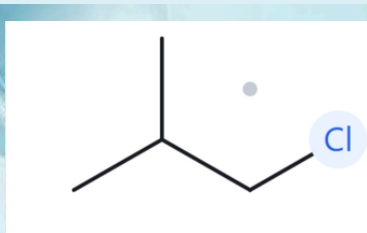
CARD B05 · NAME ➡️ STRUCTURE

FOUNDATION ⚫ 1-chloro-2-methylpropane

1. Name Decoded

Parent: Propane → a 3-carbon chain.
Substituents: Chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 carbons; open chain → C_4H_{10}. One hydrogen is replaced by chlorine → C_4H_9Cl

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_4H_9Cl.
Open-chain formula: C_4H_{10}; replacing one hydrogen with chlorine gives C_4H_9Cl. ✓

4. Back-Check

Returned by The Alkane: 1-chloro-2-methylpropane.

✓ **The returned name matches the given name.**

✔️Parent Length Matches ✔️Substituent Position Match ✔️Formula Agrees ✔️Back-Check Recorded

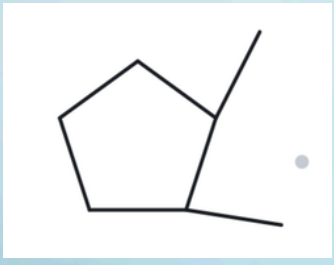
CARD B06 · NAME ➡️ STRUCTURE

FOUNDATION ⚫ 1,2 -dimethylcyclopentane

1. Name Decoded

Counted from the structure: C_4H_9Cl.
Open-chain formula: C_4H_{10}; replacing one hydrogen with chlorine gives C_4H_9Cl. ✓

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_7H_{14}.
One-ring formula: $C_nH_{2n} = C_7H_{14}$. ✓

4. Back-Check

Returned by The Alkane: 1,2-dimethylcyclopentane.

✓ **The returned name matches the given name**

✔️Parent Length Matches ✔️Substituent Position Match ✔️Formula Agrees ✔️Back-Check Recorded

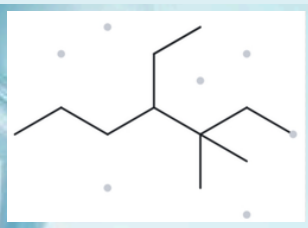
CARD B07 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 4-ethyl-3,3-dimethylheptane

1. Name Decoded

Parent: Heptane → a 7-carbon chain.
Substituents: Ethyl on C4; two methyl groups on C3.
Expected formula: 7 + 2 + 1 + 1 = 11 carbons; open chain → $C_{11}H_{24}$.

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: $C_{11}H_{24}$.
Open-chain formula: C_nH_{2n+2} = $C_{11}H_{24}$. ✓

4. Back-Check

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane.

✓ **The returned name matches the given name.**

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

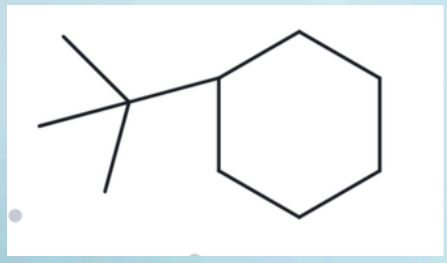
CARD B08 · NAME ➡ STRUCTURE

FOUNDATION ⚫ tert-butylcyclohexane

1. Name Decoded

Parent: Cyclohexane → a 6-carbon ring.
Substituents: tert-Butyl on C1.
Expected formula: 6 + 4 = 10 carbons; one ring → $C_{10}H_{20}$.

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: $C_{10}H_{20}$.
One-ring formula: C_nH_{2n} = $C_{10}H_{20}$. ✓

4. Back-Check

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane.

✖**mismatch:**

The Alkane returned a systematic name using the substituent name "1,1-dimethylethyl" instead of the commonly retained name "tert-butyl." Both names describe the same substituent and molecular structure. Therefore, the given name, tert-butylcyclohexane, represents the intended structure.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

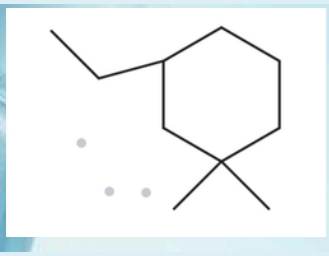
CARD B09 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 3-ethyl-1,1-dimenthylclohexane

1. Name Decoded

Parent: Cyclohexane → a 6-carbon ring.
Substituents: Ethyl on C3; two methyl groups on C1.
Expected formula: 6 + 2 + 1 + 1 = 10 carbons; one ring → $C_{10}H_{20}$.

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: $C_{10}H_{20}$.
One-ring formula: C_nH_{2n} = $C_{10}H_{20}$. ✓

4. Back-Check

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane.

✓ The returned name matches the given name.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded

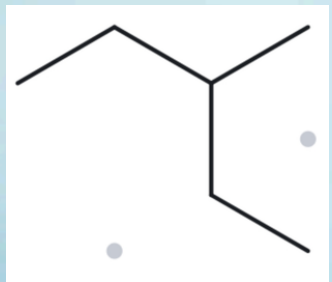
CARD B10 · NAME ➡ STRUCTURE

FOUNDATION ⚫ 2-ethylbutane

1. Name Decoded

Parent: Butane → a 4-carbon chain.
Substituents: Ethyl on C2.
Expected formula: 4 + 2 = 6 carbons; open chain → C_6H_{14}.

2. Bond-line Structure (The Alkane)



3. Formuka Check

Counted from the structure: C_6H_{14}.
Open-chain formula: C_nH_{2n+2} = C_6H_{14}. ✓

4. Back-Check

Returned by The Alkane: 3-methylpentane.

✖mismatch:
The Alkane returned 3-methylpentane because the structure contains a continuous chain of five carbon atoms, which is longer than the four-carbon butane chain in the given name. According to IUPAC naming rules, the longest continuous carbon chain must be selected as the parent chain. Therefore, the correct IUPAC name for the drawn structure is 3-methylpentane, not 2-ethylbutane.

✔Parent Length Matches ✔Substituent Position Match ✔Formula Agrees ✔Back-Check Recorded